Year 13 Further Maths - Differential equations

Similar questions - Coupled and second-order equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Two stalls are visited by P and C customers (in hundreds) at time t hours. The model is $\frac{dP}{dt} = \alpha(P-C) + \gamma t$ and $\frac{dC}{dt} = \alpha(C-P)$, where α and γ are constants.

(a) Explain why α must be negative. **[1]**

(b) Given $\alpha = -3$ and $\gamma = 18$, show that $\frac{d^2P}{dt^2} + 6\frac{dP}{dt} = 54t + 18$. **[3]**

(c) Find the complementary function, and explain why a particular integral of the form $P = at + b$ cannot work. **[3]**

(d) Using a particular integral of the form $P = at^2 + bt$, find the general solution. **[3]**

(e) When $t = 0$, $P = 40$ and $\frac{dP}{dt} = 15$. Find P when $t = 0.5$. **[4]**

(f) Explain why the model is unreliable for large t. **[1]**

2 Solve the differential equation $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 24t + 10$, given that $x = 2$ and $\frac{dx}{dt} = -1$ when $t = 0$. **[8]**

Question 1

(a) $\dfrac{\mathrm{d}C}{\mathrm{d}t} = \alpha(C-P)$: if $C > P$ the number at the busier stall should fall, so the rate must be negative when $C-P>0$, giving $\alpha < 0$

(b) $\dfrac{\mathrm{d}^2P}{\mathrm{d}t^2} = \alpha\left(\dfrac{\mathrm{d}P}{\mathrm{d}t} - \dfrac{\mathrm{d}C}{\mathrm{d}t}\right) + \gamma$. From the first equation $\alpha(P-C) = P' - \gamma t$, so $\dfrac{\mathrm{d}C}{\mathrm{d}t} = -(P'-\gamma t)$. Then $P'' = \alpha(2P' - \gamma t) + \gamma = -6P' + 54t + 18$, so $P'' + 6P' = 54t + 18$

(c) Auxiliary equation $m^2 + 6m = 0$, so $m = 0, -6$: $P = A + B\mathrm{e}^{-6t}$. A constant ($A$) is already in the complementary function, so $at+b$ would give $6a = 54t + 18$, which cannot hold for all t

(d) $P = at^2 + bt$: $2a + 6(2at + b) = 54t + 18$, so $12a = 54$ ($a = 4.5$) and $2a + 6b = 18$ ($b = 1.5$). $P = A + B\mathrm{e}^{-6t} + 4.5t^2 + 1.5t$

(e) $P' = -6B\mathrm{e}^{-6t} + 9t + 1.5$. $t = 0$: $15 = -6B + 1.5$ so $B = -2.25$ and $A = 40 + 2.25 = 42.25$. $P(0.5) = 42.25 - 2.25\mathrm{e}^{-3} + 1.125 + 0.75 = 44.01$ hundred

(f) P grows like $4.5t^2$ without bound, but the number of customers is limited by the size of the stall/population

Question 2

Auxiliary: $m^2 + 6m + 8 = 0$, $m = -2, -4$. Complementary function $c_1\mathrm{e}^{-2t} + c_2\mathrm{e}^{-4t}$. Particular integral $x = at + b$: $6a + 8at + 8b = 24t + 10$ so $a = 3$, $b = -1$. General solution $x = c_1\mathrm{e}^{-2t} + c_2\mathrm{e}^{-4t} + 3t - 1$. $x(0) = 2$: $c_1 + c_2 = 3$. $x'(0) = -2c_1 - 4c_2 + 3 = -1$: $c_1 + 2c_2 = 2$. So $c_2 = -1$, $c_1 = 4$. $x = 4\mathrm{e}^{-2t} - \mathrm{e}^{-4t} + 3t - 1$